Practical 6: Configuring a Zone-Based Policy Firewall (ZPF)

t configure R1

```
enable
configure terminal
hostname R1
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
interface s0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 64000
no shutdown
line console 0
password conpa55
login
line vty 0 4
password vtypa55
login
enable secret enpa55
ip domain-name ccnasecurity.com
username admin secret adminpa55
crypto key generate rsa
1024
line vty 0 4
login local
transport input ssh
ip route 192.168.3.0 255.255.255.0 10.1.1.2
end
write memory
```

Now configure R2

```
enable
configure terminal
hostname R2
interface s0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
interface s0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 64000
```

```
no shutdown
line console 0
password conpa55
login
line vty 0 4
password vtypa55
login
enable secret enpa55
ip domain-name ccnasecurity.com
username admin secret adminpa55
crypto key generate rsa
1024
line vty 0 4
login local
transport input ssh
ip route 192.168.1.0 255.255.255.0 10.1.1.1
ip route 192.168.3.0 255.255.255.0 10.2.2.1
end

write memory
```

configure R3 (IMPORTANT – ZPF here)

```
enable
configure terminal
hostname R3
interface g0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
interface s0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
line console 0
password conpa55
login
line vty 0 4
password vtypa55
login
enable secret enpa55
ip domain-name ccnasecurity.com
username admin secret adminpa55
```

```
crypto key generate rsa
1024
line vty 0 4
login local
transport input ssh
ip route 192.168.1.0 255.255.255.0 10.2.2.2
! ========================
! ZONE BASED FIREWALL
! ========================
zone security IN-ZONE
zone security OUT-ZONE
access-list 101 permit ip 192.168.3.0 0.0.0.255 any
class-map type inspect match-all IN-NET-CLASS-MAP
match access-group 101
policy-map type inspect IN-2-OUT-PMAP
class type inspect IN-NET-CLASS-MAP
inspect
zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE
service-policy type inspect IN-2-OUT-PMAP
interface g0/0
zone-member security IN-ZONE
interface s0/1/0
zone-member security OUT-ZONE
end
write memory
```

Practical 7: Layer 2 Security

🎯 Aim

To configure Layer 2 security on switches by:

- Assigning a root bridge
- Securing Spanning Tree Protocol (STP)
- Enabling port security to prevent unauthorized access and CAM table attacks

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🖥️ R1 (Router Configuration)

Enable

configure terminal

```
enable secret enpa55
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
exit
crypto key generate rsa
1024
end
```

🖧 Central Switch (Primary Root Bridge)

```
enable
configure terminal
enable secret enpa55
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
```

```
exit
crypto key generate rsa
1024
spanning-tree vlan 1 root primary
end
```

🖧 SW1 (Secondary Root + Security)

```
enable
configure terminal
enable secret enpa55
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
exit
crypto key generate rsa
1024
spanning-tree vlan 1 root secondary
interface range fa0/1 - 4
spanning-tree portfast
spanning-tree bpduguard enable
exit
interface range fa0/23 - 24
```

```
spanning-tree guard root
exit
interface range fa0/1 - 22
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
exit
interface range fa0/5 - 22
shutdown
end
```

🖧 SW2 (Layer 2 Security Configuration)

```
enable
configure terminal
enable secret enpa55
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
exit
crypto key generate rsa
```

1024

interface range fa0/1 - 4

spanning-tree portfast

spanning-tree bpduguard enable

exit

interface range fa0/23 - 24

spanning-tree guard root

exit

interface range fa0/1 - 22

switchport mode access

switchport port-security

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security mac-address sticky

exit

interface range fa0/5 - 22

shutdown

end

✅ Verification Commands

show spanning-tree

show port-security interface fa0/1

Practical 8: Layer 2 VLAN Securit

R1

enable

configure terminal

enable secret enpa55

```
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
exit
crypto key generate rsa
1024
interface gig0/0.1
encapsulation dot1q 5
ip address 192.168.5.100 255.255.255.0
interface gig0/0.2
encapsulation dot1q 10
ip address 192.168.10.100 255.255.255.0
interface gig0/0.15
encapsulation dot1q 15
ip address 192.168.15.100 255.255.255.0
interface gig0/0.3
encapsulation dot1q 20
ip address 192.168.20.100 255.255.255.0
access-list 101 deny ip any 192.168.20.0 0.0.0.255
access-list 101 permit ip any any
access-list 102 permit ip host 192.168.20.50 any
```

```
interface gig0/0.1
ip access-group 101 in
interface gig0/0.2
ip access-group 101 in
line vty 0 4
access-class 102 in
end
```

SWA (Access Switch A)

```
enable
configure terminal
enable secret enpa55
vlan 5
exit
vlan 10
exit
vlan 15
exit
vlan 20
exit
interface fa0/2
switchport mode access
switchport access vlan 10
interface fa0/3
switchport mode access
switchport access vlan 10

interface fa0/4
switchport mode access
```

switchport access vlan 5

interface fa0/1

switchport mode access

switchport access vlan 20

interface fa0/24

switchport mode trunk

switchport trunk native vlan 15

interface vlan 20

ip address 192.168.20.1 255.255.255.0SWB (Access Switch B)

enable

configure terminal

enable secret enpa55

vlan 5

exit

vlan 10

exit

vlan 15

exit

vlan 20

exit

interface fa0/1

switchport mode access

switchport access vlan 5

interface fa0/2

switchport mode access

switchport access vlan 5

interface fa0/3

switchport mode access

```
switchport access vlan 5
interface fa0/4
switchport mode access
switchport access vlan 10
interface fa0/24
switchport mode trunk
switchport trunk native vlan 15
interface vlan 20
ip address 192.168.20.2 255.255.255.0
end
end
SW1
enable
configure terminal
enable secret enpa55
vlan 5
exit
vlan 10
exit
vlan 15
exit
vlan 20
exit
interface fa0/24
switchport mode trunk
switchport trunk native vlan 15
interface gig0/1
switchport mode trunk
```

```
switchport trunk native vlan 15
interface fa0/23
switchport mode trunk
switchport trunk native vlan 15
switchport nonegotiate
interface vlan 20
ip address 192.168.20.3 255.255.255.0
end
```

🖧 SW2

```
enable
configure terminal
enable secret enpa55
vlan 5
exit
vlan 10
exit
vlan 15
exit
vlan 20
exit
interface fa0/24
switchport mode trunk
switchport trunk native vlan 15
interface gig0/1
switchport mode trunk
switchport trunk native vlan 15
interface fa0/23
```

```
switchport mode trunk
switchport trunk native vlan 15
switchport nonegotiate
interface vlan 20
ip address 192.168.20.4 255.255.255.0
end
```

🖧 Central Switch

```
enable
configure terminal
enable secret enpa55
vlan 5
exit
vlan 10
exit
vlan 15
exit
vlan 20
exit
interface range gig0/1-2
switchport mode trunk
switchport trunk native vlan 15
interface fa0/1
switchport mode trunk
switchport trunk native vlan 15
interface vlan 20
ip address 192.168.20.5 255.255.255.0
end
```

✅ Verification Commands

show vlan brief

show interfaces trunk

Practical 9: Configure IOS Intrusion Prevention System

R1 (Main IPS Router Configuration)

enable

configure terminal

enable secret enpa55

line console 0

password conpa55

login

exit

ip domain-name ccnasecurity.com

username admin secret adminpa55

line vty 0 4

login local

transport input ssh

exit

crypto key generate rsa

1024

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.1.1.0 0.0.0.3 area 0

end

🖥️ R2 (OSPF Configuration)

enable

```
configure terminal
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
end
```

🖥️ R3 (OSPF Configuration)

```
enable
configure terminal
router ospf 1
network 10.2.2.0 0.0.0.3 area 0
network 192.168.3.0 0.0.0.255 area 0
end
```

🛡️ Enable IOS IPS (On R1)

```
enable
license boot module c1900 technology-package securityk9
yes
copy running-config startup-config
reload
```

📂 IPS Configuration (On R1)

```
enable
configure terminal
ip ips config location flash:ipsdir
ip ips name iosips
ip ips notify log
service timestamps log datetime msec
```

```
logging host 192.168.1.50
ip ips signature-category
category all
retired true
exit
category ios_ips basic
retired false
exit
exit
interface gig0/0
ip ips iosips out
end
```

🛠️ Create IPS Directory (Run in Privileged Mode)

```
mkdir ipsdir
```

🔧 Modify IPS Signature (On R1)

```
configure terminal
ip ips signature-definition
signature 2004 0
status
retired false
enabled true
exit
engine
event-action produce-alert
event-action deny-packet-inline
exit
```

```
exit
exit
end
```

✅ Verification Commands

```
show ip ips all
```